

Reading Response: Exploring Sound Awareness in the Home for People who are Deaf or Hard of Hearing – Jain et al.

Summary: This study investigates how Deaf and hard-of-hearing (DHH) individuals perceive and manage sounds at home, using this insight to design and test three sound awareness prototypes (a list, a floor plan, and a waveform). The researchers identified that participants had a strong preference for the floorplan design because of its intuitive spatial visualization and room specificity. Still, the list format was also valued for providing a historical record of home noises, which could potentially help participants put sounds in context. A strength of the paper and a section of the paper I found particularly unique/interesting was the establishment of a “design space” early in the process (made me think of set theory, linear algebra, domains, etc), yet the study was limited by a small sample size where most participants lived alone, a limitation because this could potentially fail to capture the dynamics of having roommates. My main questions revolve around the methodology and participant motivations. I am unsure what the specific criteria for "thematic saturation" are, and I didn't find that term to be well-defined. I also wonder what the design process was for selecting the specific thematic scenarios used in the Wizard-of-Oz part of the study. Finally, the paper mentions a preference for form factors that are not easily recognized as assistive technology... I wonder why invisibility is prioritized for participants; is this an aesthetic choice, or is it an effort to avoid social stigma or burdening others around you?

Reaction: One thing I found compelling about this study was how ease of use seemed to dictate the entire study; the researchers prioritized glanceability and intuitive visualizations over complex data (supported by the lack of support for the waveform prototype), which resonated with participants' desire for immediate, actionable information. The varying preferences for notifications—specifically the distinction between "need to know" actionable sounds versus background noise—were also thought-provoking. I found this interesting because in our Group Project, I am wondering how to choose notification frequency and style. To apply these findings to inclusive technology design, I would focus on the loudness and cost-effectiveness aspects mentioned by participants. For example, since loudness perception can be subjective or hard to visualize, I would design a sort of calibration step where a trusted hearing person helps the DHH user define what constitutes "loud" versus "soft" in their specific environment.